

SWAGATI PACHARE

Data Scientist | AI/ML Engineer

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PROFESSIONAL SUMMARY

Data Scientist and Python Developer with hands-on experience building end-to-end machine learning and deep learning pipelines. Skilled in EDA, predictive modeling, computer vision, NLP, and Power BI dashboards. Experienced across the full data science lifecycle — data collection, feature engineering, model training, evaluation, and deployment — achieving 80-92% accuracy across real-world projects.

TECHNICAL SKILLS

Programming: Python, SQL

Data Science: EDA, Statistical Analysis, Data Preprocessing, Feature Engineering, Predictive Modeling, A/B Testing, Data Wrangling

Machine Learning: Supervised & Unsupervised Learning, Classification, Regression, Clustering, SVM, Random Forest, Cross-Validation, Hyperparameter Tuning

Deep Learning & CV: CNN, LSTM, Neural Networks, YOLOv5, Object Detection, OpenCV, TensorFlow, Keras

NLP & Audio: MFCC Feature Extraction, Speech Processing, Text Classification

Data Visualization: Power BI, DAX, KPI Development, Dashboard Design, Matplotlib, Seaborn

Libraries & Frameworks: NumPy, Pandas, Scikit-learn, Flask, Django

Databases & Tools: MySQL, GitHub, Jupyter Notebook

WORK EXPERIENCE

Data Scientist / Software Developer, Cojag Smart Technology Pvt. Ltd. | Nagpur (*June 2025 – January 2026*)

- Designed and deployed Python-based data processing pipelines for analytics dashboards, automating reporting workflows and reducing manual data handling
- Integrated ML and deep learning models into production pipelines; optimized inference performance through code profiling and refactoring
- Performed EDA and feature engineering to prepare datasets for model training, directly improving downstream model accuracy
- Managed MySQL schema design and query optimization for real-time data retrieval powering ML feature stores and dashboards

Software Developer Intern, Cojag Smart Technology Pvt. Ltd. | Nagpur (*October 2024 – May 2025*)

- Built interactive client-facing dashboards using Flask, HTML, and CSS to monitor data trends in real time
- Implemented ML processing scripts for data preprocessing, model inference, and result visualization
- Supported end-to-end model lifecycle tasks including data ingestion, evaluation, and QA to ensure model reliability in production

PROJECTS

Pothole Detection System — Computer Vision & YOLOv5

- Built a real-time detection system using YOLOv5 and OpenCV achieving 92% accuracy at 25 FPS on live video feeds
- Engineered a full data pipeline (collection, annotation, augmentation, preprocessing) and integrated MySQL for detection event logging and geospatial analytics

HR Analytics Dashboard — Power BI

- Developed an interactive HR Analytics Dashboard analyzing attrition across age, salary, education, and job role dimensions
- Authored DAX measures and KPIs; identified higher attrition in younger, lower-salary groups to drive data-driven HR strategy

Speech Emotion Recognition — Machine Learning & Deep Learning

- Extracted MFCC audio features from raw speech data for multi-class emotion classification
- Trained and benchmarked CNN, LSTM, SVM, and Random Forest models; achieved 80% accuracy with k-fold cross-validation and hyperparameter tuning

EDUCATION

Master of Computer Science (MCA), RTMNU University, Nagpur | CGPA: 8.5/10 (*2023 – 2025*)

Bachelor of Computer Science (BCA), RTMNU University, Nagpur | CGPA: 8.0/10 (*2020 – 2023*)

CERTIFICATIONS

GenAI Powered Data Analytics Job Simulation — Tata (via Forage) | *June 2026*

Data Analytics Job Simulation — Deloitte (via Forage) | *April – June 2026*